Deepfake detection using Deep Learning

Subject Code: CM461

Batch No :-
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REFERENCES

INTRODUCTION

- Rapid advancements in **Artificial Intelligence (AI), Machine Learning, and Deep Learning** have led to powerful tools for manipulating multimedia content.
- These manipulations, known as **deepfakes**, are highly realistic and difficult for humans to detect.
- With the development of **Generative Adversarial Networks (GANs)** and diffusion models, deepfake content has become more advanced and widespread.
- Deepfakes pose serious threats to:
  - Digital security
  - Information integrity
  - Social media trust
  - Personal privacy
  - Democratic processes



PROBLEM STATEMENT

- The increasing availability of deepfake tools has made it easy to create highly realistic fake content.
- Existing detection methods struggle to identify **high-quality deepfakes**.
- Traditional machine learning approaches rely on **manual feature engineering**, while deep learning models often require **high computational resources**.
- Deepfakes look almost real, making them hard to detect.
- There is a need for a system that:
  - Provides **high accuracy**
  - Works for both **images and videos**
- **Project Goal:**

  To develop an intelligent system that can accurately distinguish between **real and deepfake images and videos**.

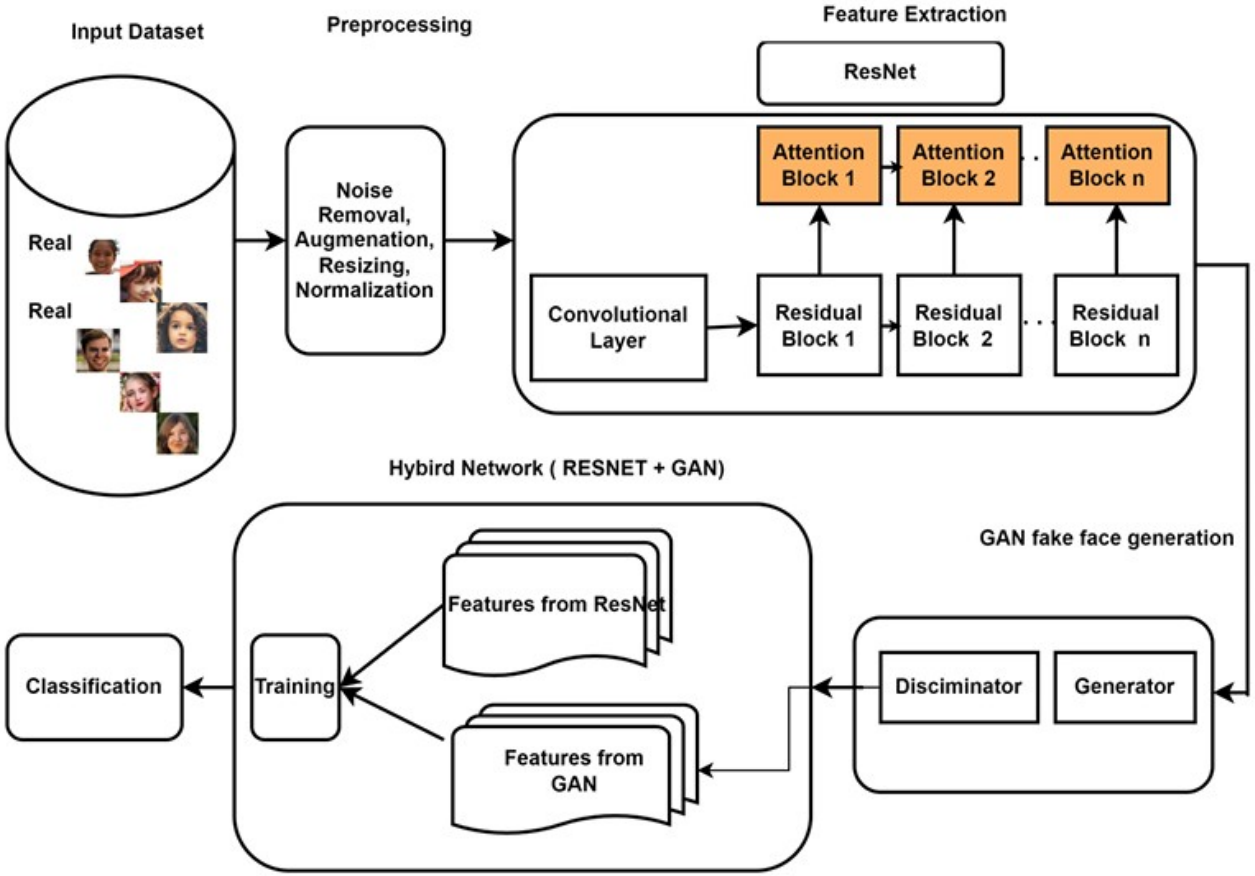
LITERATURE SURVEY

S.No	Method/Model	Dataset Used	Technique Used	Accuracy
1	SVM	Real & Fake Face Dataset	Handcrafted Features	65%
2	Random Forest	Real & Fake Face Dataset	Statistical Feature-Based	70%
3	CNN (Basic)	FaceForensics++	Convolutional Neural Network	75%
4	VGG16	Celeb-DF	Deep CNN	78%
5	ResNet-50	Real & Fake Face Dataset	Residual Learning	80%
6	GAN + ResNet Hybrid	Real & Fake Face Dataset (2041 Images)	Generative + CNN Hybrid	83%

RESEARCH GAP

Existing Models (Literature)	Research Gap (Identified Problem)
Use traditional ML (SVM, RF) with handcrafted features	Low accuracy and poor feature representation
CNN models (VGG, ResNet) focus on spatial features	Lack of temporal feature learning for videos
ResNet-based models are computationally heavy	Need for lightweight and efficient architectures
GAN-based hybrid models improve accuracy	Training instability and high complexity
Models detect only image-based deepfakes	No unified system for both images and videos
Limited focus on important regions (no attention)	Need for attention mechanisms to detect subtle artifacts

EXISTING WORKFLOW DIAGRAM



- The existing system uses a hybrid model combining GAN and ResNet for deepfake detection.
- ResNet50 is used to extract deep spatial and texture features from images.
- GAN generates realistic fake images and helps improve detection capability.
- The discriminator learns to distinguish between real and fake faces.

PROJECT OBJECTIVES

- To develop a highly accurate deepfake detection system.
- To detect both fake images and fake videos.
- To use EfficientNet models for parameter-efficient feature extraction.
- To apply attention mechanisms (CBAM & Temporal Attention) for better feature focus.
- To capture both spatial and temporal features using deep learning models.
- To improve overall detection performance compared to existing models

DATASET USED

- **Image Dataset:** 140k Real and Fake Faces with ELA (Error Level Analysis).

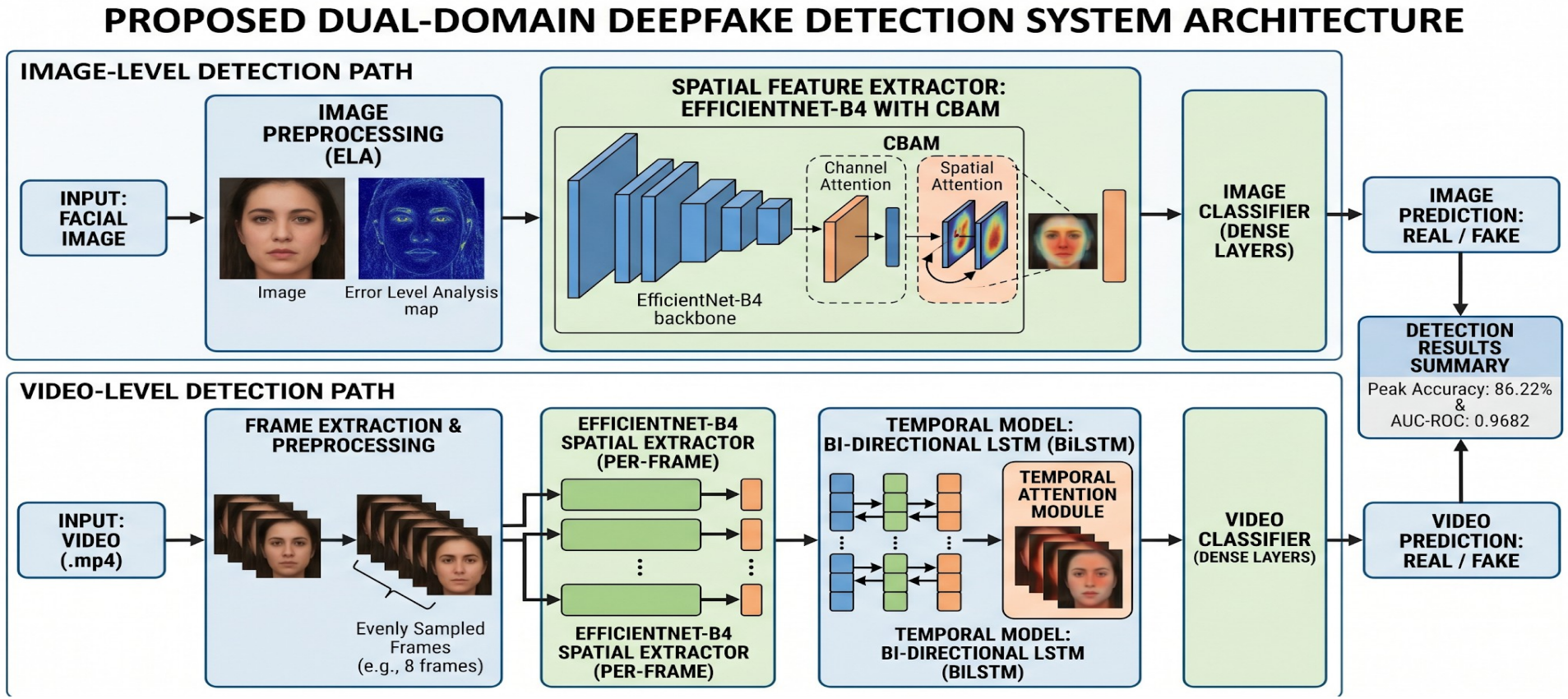
Split	Total Samples	Fake Samples	Real Samples
Training	100,000	50,000	50,000
Validation	20,000	10,000	10,000
Testing	20,000	10,000	10,000
Total	1,40,000	70,000	70,000

- **Video Dataset:** Celeb-DF-v2 (Sequence Analysis).

Split	Total Videos	Fake Videos	Real Videos
Training	4,808	4,238	570
Validation	1,203	1,061	142
Testing	518	-	-
Total	6,011	5,299	712

- **Data Breakdown:** 5,299 fake videos (Celeb-synthesis) and 712 real videos (Celeb-real & YouTube-real).

PROPOSED MODEL



Approach

A **dual-domain deepfake detection system** is proposed for both **images and videos**.
Uses **EfficientNet with attention mechanisms** instead of heavy ResNet models.

Image-Based Detection

Model: **EfficientNet-B4 + CBAM**
Steps:

- Image preprocessing (ELA)
- Feature extraction using EfficientNet-B4
- CBAM highlights important regions
- Dense layers
- Real/Fake classification

Video-Based Detection

Model: **EfficientNet-B0 + BiLSTM + Temporal Attention**
Steps:

- Extract frames from video
- Feature extraction (EfficientNet-B0)
- Sequence learning using BiLSTM
- Temporal attention selects key frames
- Final classification

PROPOSED SYSTEM WORKFLOW

- **Data Collection**
  Real and Fake Face dataset containing real images, deepfake images, and videos.
- **Pre-Processing**
  Image resizing, normalization, data augmentation, and frame extraction for videos.
- **Feature Extraction**
  EfficientNet-B4 extracts high-level spatial features from images.
  EfficientNet-B0 extracts frame-level features from video sequences.
- **Attention Mechanisms**
  CBAM highlights important spatial and channel features in images.
  Temporal Attention selects important frames in videos.

- **Model Selection**
  Hybrid deep learning model using EfficientNet + Attention + BiLSTM.
- **Train–Test Split and Model Fitting**
  Dataset split into training and testing sets.
  Model trained using binary cross-entropy loss and Adam optimizer.
- **Temporal Modeling (Video)**
  BiLSTM captures temporal dependencies between video frames.
- **Model Evaluation and Deployment**
  Performance evaluated using accuracy, precision, recall, and F1-score.
  Trained model used for real-time deepfake detection of images and videos.

TOOLS & TECHNOLOGIES

Programming Language: Python

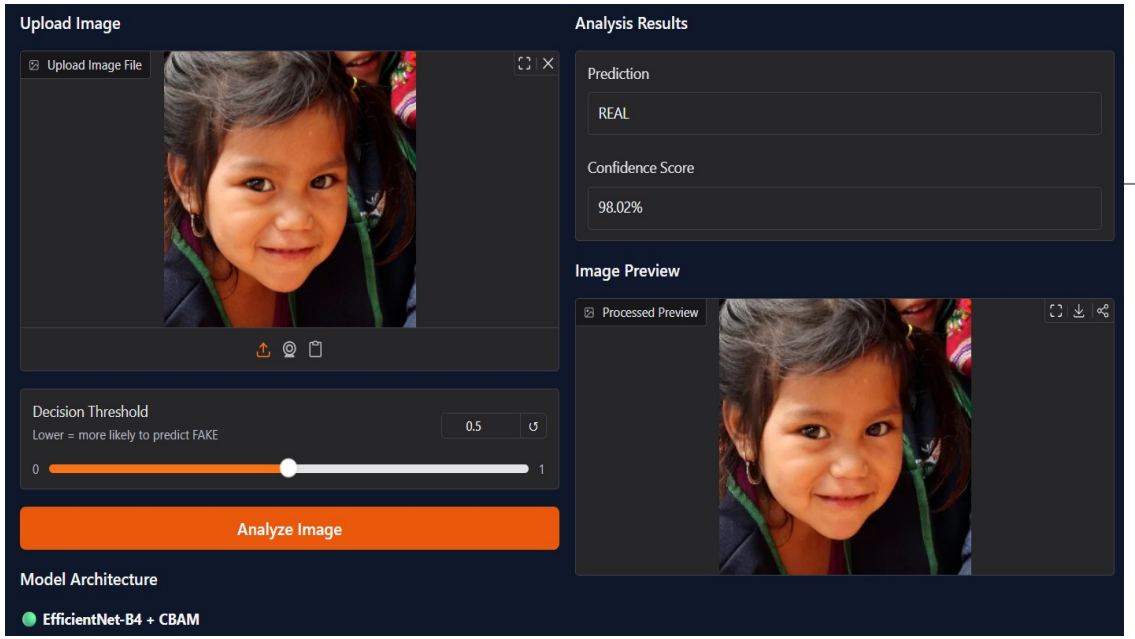
Deep Learning Frameworks: TensorFlow/Keras, PyTorch

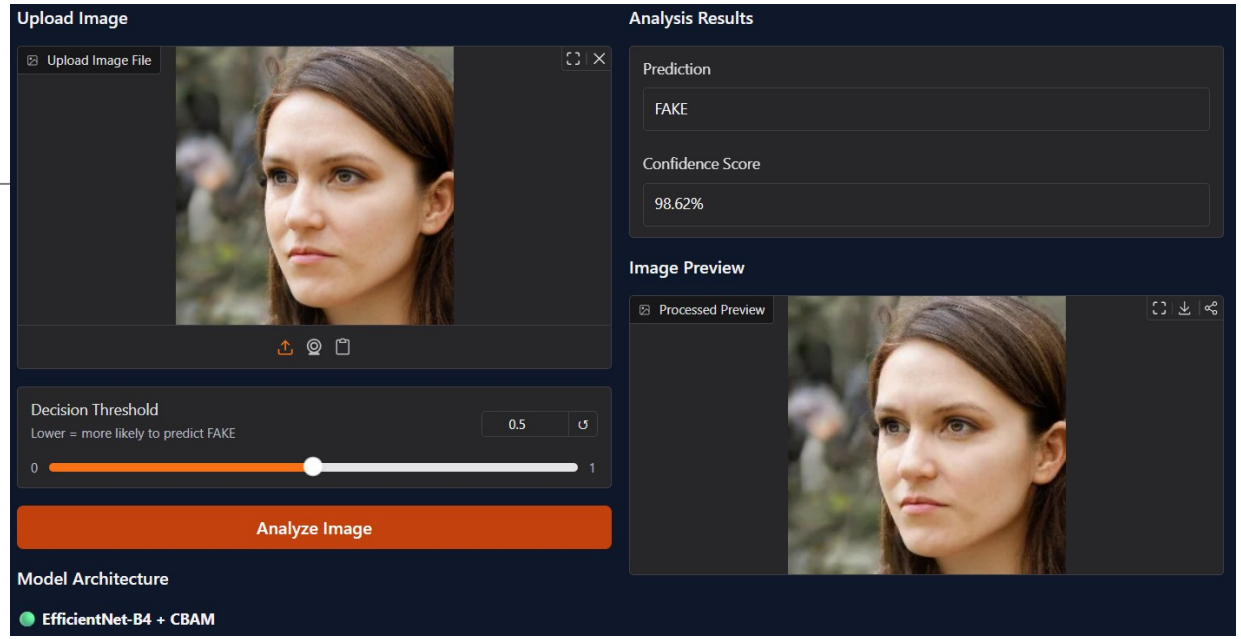
Data Processing & Visualization: Pandas, NumPy, Matplotlib, Seaborn, PIL

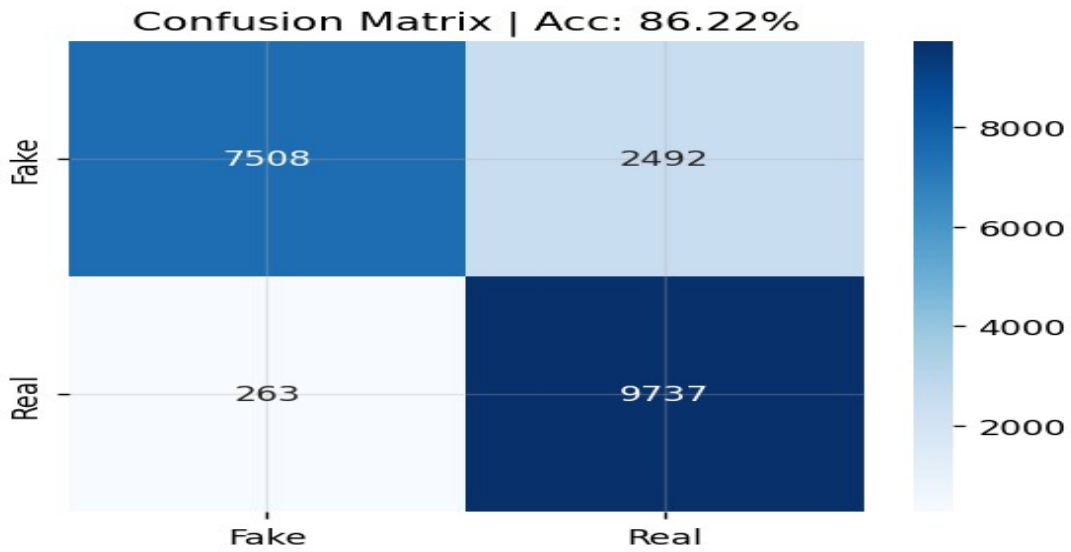
Hardware Accelerator: Dual Kaggle GPUs (Tesla T4 x2 / Tesla P100)

Deployment & Hosting: Hugging Face Spaces (for building a web-based interactive interface).

RESULTS AND ANALYSIS







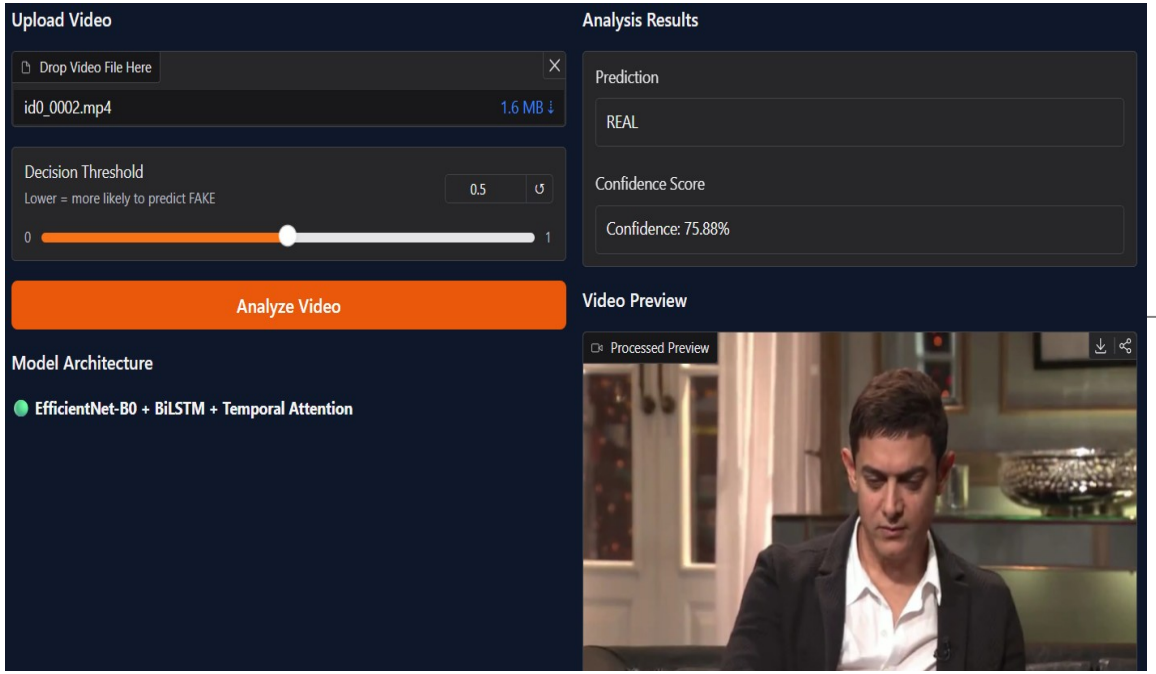
🎯 FINAL METRICS

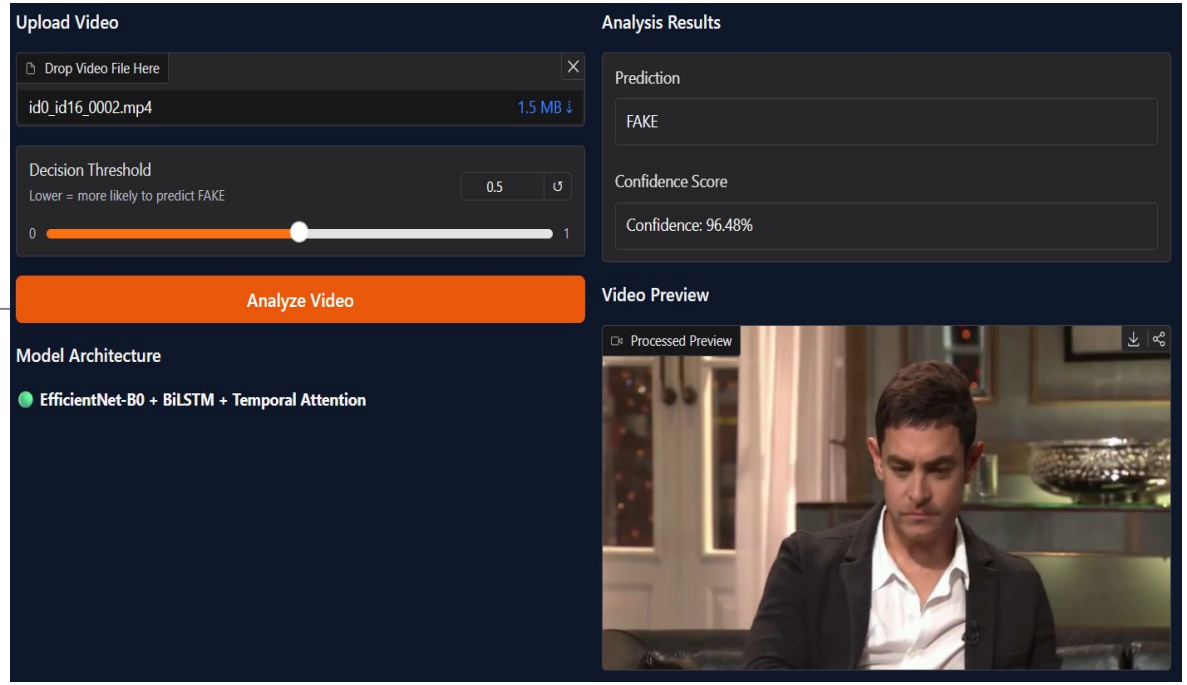
Test Accuracy : 86.22%

AUC-ROC : 0.9682

Average Precision : 0.9681

	precision	recall	f1-score	support
Fake	0.9662	0.7508	0.8450	10000
Real	0.7962	0.9737	0.8761	10000
accuracy			0.8622	20000
macro avg	0.8812	0.8622	0.8605	20000
weighted avg	0.8812	0.8622	0.8605	20000





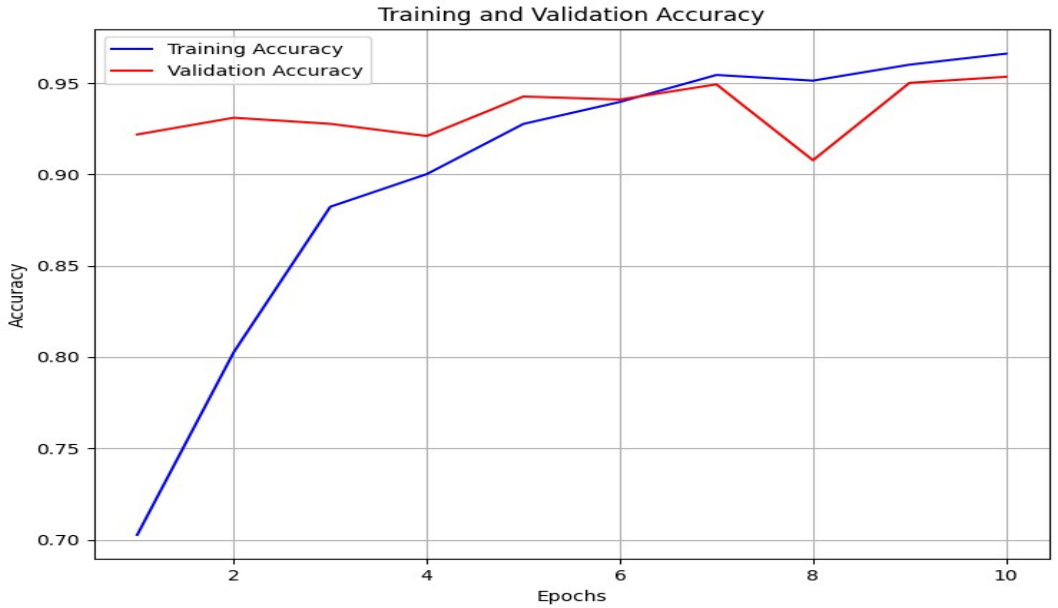

```
Epoch 10/10
  Train Loss: 0.0088 | Train Acc: 0.9661
  Val Loss:   0.0179 | Val Acc:   0.9534
  Precision:  0.9057 | Recall:    0.6761 | AUC: 0.9668
  Confusion Matrix:
[[1051   10]
 [  46   96]]
------------------------------------------------------------
Validation accuracy improved from 0.9501 to 0.9534. Saving model...
Saving checkpoint at epoch 10...

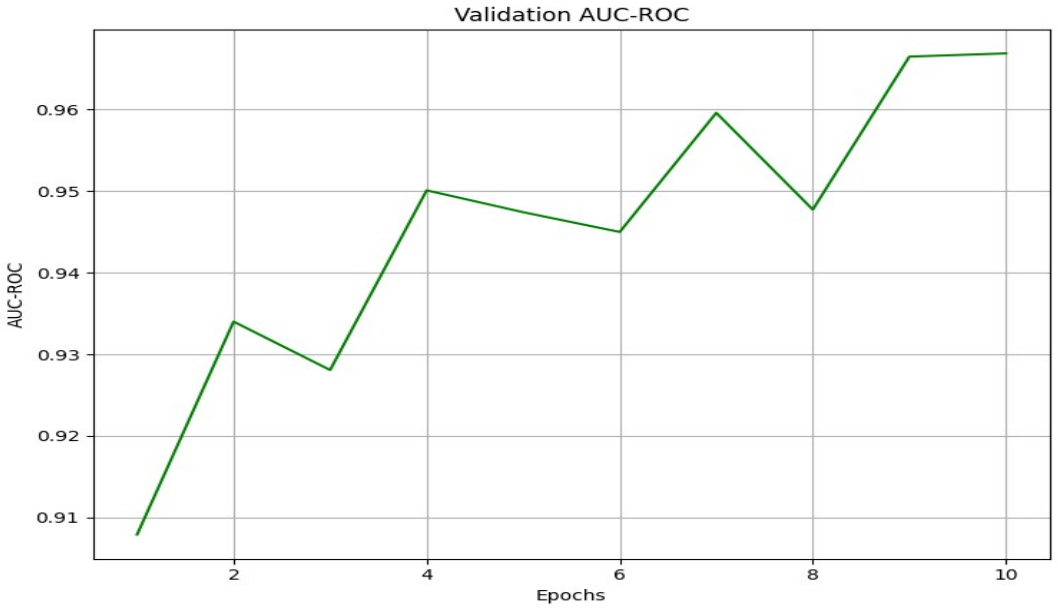
Epoch 10/10: 100%|██████████| 10/10 [3:35:57<00:00, 1295.72s/epoch]

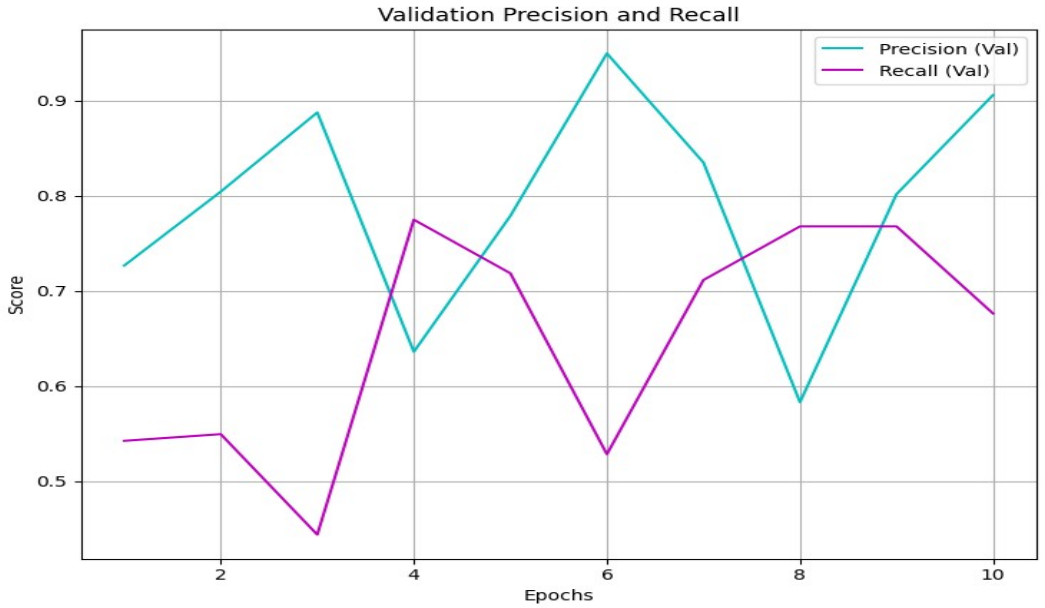
Best Validation Accuracy: 0.9534
```

View


Training and Validation Loss
Training Loss
Validation Loss
Loss
Epochs
0.05
0.04
0.03
0.02
0.01
2
4
6
8
10


Training and Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epochs
0.95
0.90
0.85
0.80
0.75
0.70
2
4
6
8
10


Validation AUC-ROC
AUC-ROC
Epochs
0.96
0.95
0.94
0.93
0.92
0.91
2
4
6
8
10


Validation Precision and Recall
Precision (Val)
Recall (Val)
Score
Epochs
0.9
0.8
0.7
0.6
0.5
2
4
6
8
10

CONCLUSION

- This project presents an effective deepfake detection system using **EfficientNet and attention mechanisms**.
- The model successfully captures both **spatial features (images)** and **temporal patterns (videos)**.
- Compared to traditional methods, it provides **better accuracy with lower computational cost**.
- Attention modules help in focusing on **manipulated facial regions**, improving detection performance.
- The system demonstrates strong potential for **real-world applications** like cybersecurity and media verification.
- Overall, the proposed approach offers a **robust, scalable, and efficient solution** for deepfake detection.

FUTURE SCOPE

- Develop a **real-time deepfake detection system** for instant analysis.
- Deploy the model as a **web extension or API** for social media platforms.
- Expand the framework for multimodal forgery detection, including synthetic media, generated text, and digitally manipulated identity documentsUse advanced architectures like **transformers** for improved performance.
- Optimize the model using **quantization** for deployment on mobile and edge devices.
- Expand datasets to include **audio-visual and multimodal deepfake detection** for better generalization.

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[2] S. Safwat, A. Mahmoud, I. E. Fattoh, and F. Ali, "*Hybrid Deep Learning Model Based on GAN and RESNET for Detecting Fake Faces," IEEE Access,* vol. 12, pp. 86391–86402, 2024. DOI:10.1109/ACCESS. 2024. 3416910.

[3] B. Chesney and D. Citron, "*Deep fakes: A looming challenge for privacy, democracy, and national security," Calif. L. Rev., vol. 107, p. 1753, Jan. 2019. [4] M. Tan and Q. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in Proc. Int. Conf. Machine Learning (ICML)*, 2019.

THANK YOU!

